CS & IT ENGINEERING

COMPILER DESIGN

PARSING

Lecture No.- 02

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Recap of Previous Lecture

Topic ??

Parsing

① Top down Parsing

② Bottom up Parsing

→ LR Parsing

→ Operator Precedence Parsing

Topics to be Covered

Topic ?? LR Parsing

Topic ?? LR Parsing Table Construction

Topic ??

Topic : Shift Reduce Parser in Compiler

$S \to S+S$ ✓

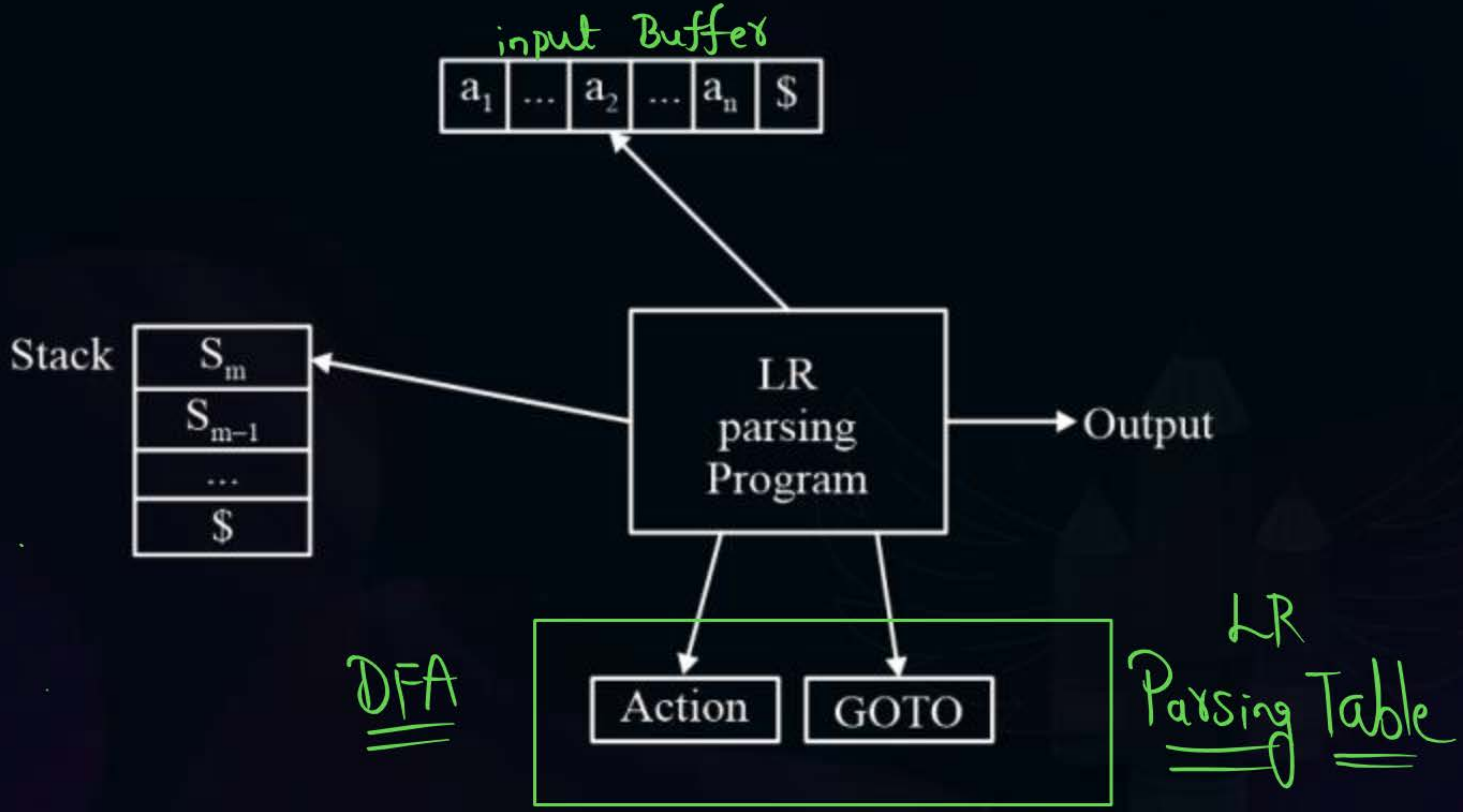
$S \to S*S$

$S \to id$

Stack	Input Buffer	Parsing Action
$	id+id+id$	Shift
$id	+ id +id$	Reduce by S --> id
$S	+ id +id$	Shift
$S+	id+id$	Shift
$ S + id	+id$	Reduce by S --> id
$ S + S	+id$	Shift
$ S + S +	id$	Shift
$ S + S + id	$	Reduce by S --> id
$ S + S + S	$	Reduce by S--> S + S
$ S + S	$	Reduce by S--> S + S
$S	$	Accept

Topic : LR Parsing

Shift / Reduce Parser



Parsing Table

- LR(0) Table
- SLR(1) Parsing Table
- LALR(1) Parsing Table
- CLR(1) Parsing Table

#Q. How many shift action and reduce action are taken by LR parser to parse input ab by using following grammar and parsing table.

$$S \rightarrow AB$$
$$A \rightarrow a$$
$$B \rightarrow b$$

State of DFA

	a	b	$	S	A	B
I_0	S_3			1	2	
I_1			Accept			
I_2		S_5				4
I_3		r_2				
I_4			R_1			
I_5			r_3			

Topic : Types of LR Parser

① LR(0) Parser

② **SLR(1) Simple LR Parser:**

- Works on smallest class of grammar
- Few number of states, hence very small table
- Simple and fast construction

③ **LR(1)-LR Parser:**

- Works on complete set of LR(1)
- Grammar Generates large table and large number of states
- Slow construction

④ **LALR(1) - Look-Ahead LR Parser:**

- Works on intermediate size of grammar
- Number of states are same as in SLR(1)

#Q. How many shift action and reduce action are taken by LR parser to parse input ab by using following grammar and parsing table.

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b$

DFA States

	A a	B b	$	S	A	B
I_0	S_3			1	2	
I_1			Accept			
I_2		S_5				4
I_3		r_2				
I_4			R_1			
I_5			r_3			

Topic : LR(0) Table Construction

1. Construct Augmented Grammar

2. Construct LR(0) items DFA by using closure() and $\boxed{\text{goto()}}$.

3. Construct parsing table

Topic : LR(0) Table Construction

$$S' \to S$$
$$S \to AB$$
$$A \to a$$
$$B \to b$$

1. Construct Augmented Grammar ($S' \to S$)

2. Construct LR(0) items DFA by using $\boxed{\text{closure()}}$ and $\boxed{\text{goto()}}$.

3. Construct parsing table

Topic : LR(0) Items

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b$

$S \rightarrow \cdot A B$

$S \rightarrow \underline{A} \cdot B$

$S \rightarrow \boxed{A B} \cdot$

Reduce

$A \rightarrow \boxed{X Y} \cdot Z$

Topic : Closure()

Closure()

$S \to AB$

$A \to a$

$B \to b$

closure()

$S \to A \cdot B$

$B \to \cdot b$

closure()

$S \to \cdot A B$

$A \to \cdot a$

$$\left\{\begin{array}{l} E \to T + E \\ E \to T \end{array}\right\}$$

$$T \to a$$

closure()

$$E' \to .E$$

3

$$\left\{\begin{array}{l} E \to .T + E \\ E \to .T \\ T \to .a \end{array}\right.$$

$S \to Aa$

$S \to Bb$

$A \to a$

$B \to b$

closure()

$S' \to \cdot S$

I_0

$S \to \cdot Aa$

$S \to \cdot Bb$

$A \to \cdot a$

$B \to \cdot b$

$S \to aAb$

$A \to a$

closure()

① $\boxed{\begin{array}{l} S' \to \cdot S \\ S \to \cdot aAb \end{array}}$

Topic : Goto()

1 $$\boxed{S \to A \cdot B} \xrightarrow{\text{goto}} \boxed{S \to AB \cdot}$$

2 $$\boxed{\begin{array}{l} E \to \cdot T + E \\ E \to \cdot T \end{array}} \xrightarrow{\text{goto}} \boxed{\begin{array}{l} E \to T \cdot + E \\ E \to T \cdot \end{array}}$$

Construct LR(0) items DFA for given grammar.

#Q. Construct LR(0) parsing table for the following grammar.

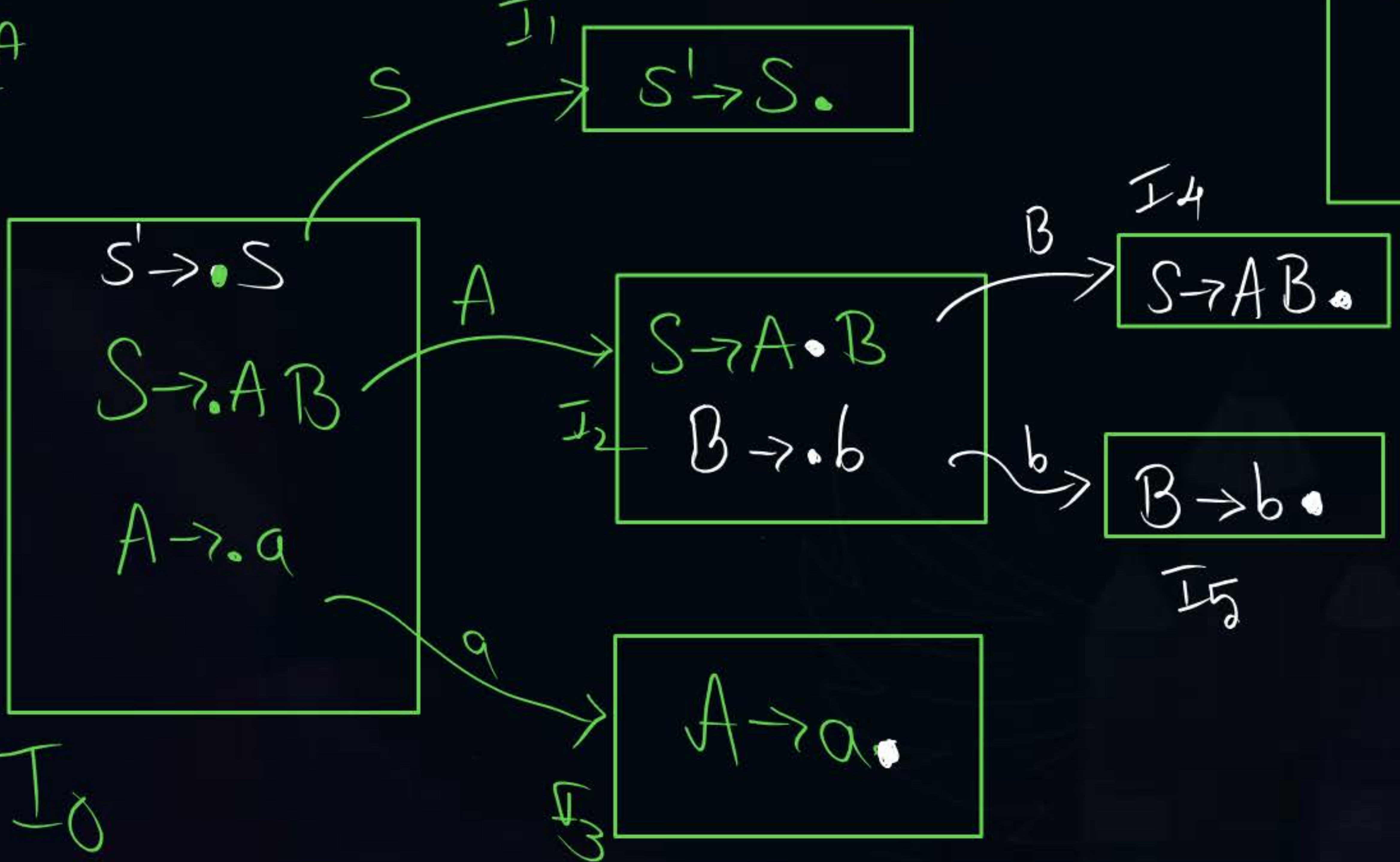
$S' \to S$

1. $S \to AB$
   $A \to a$
   $B \to b$

LR(0) items DFA

closure()

6 states



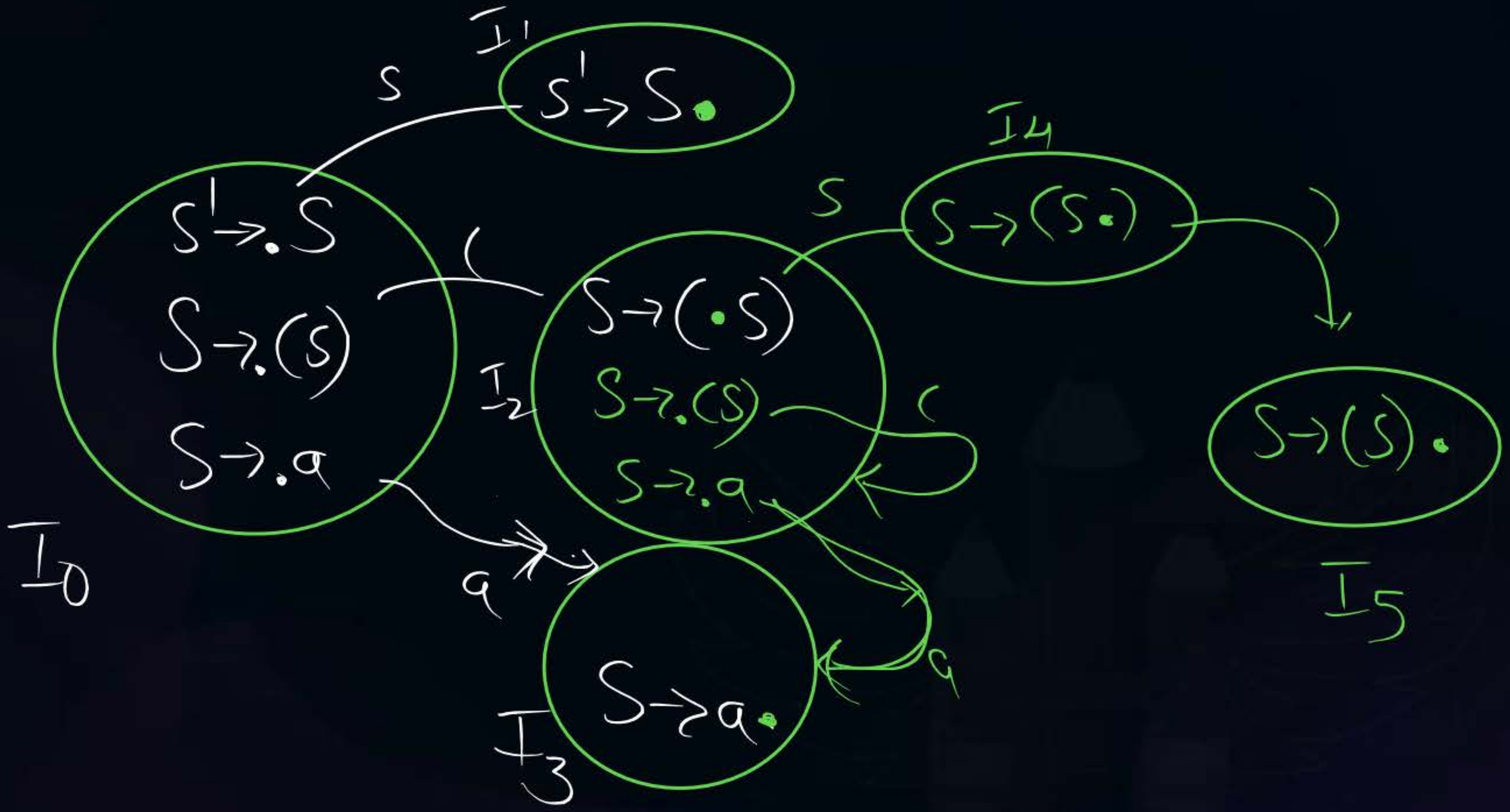
(Q) How many States present in LR(0) items DFA for given grammar.

(1)

$S' \to S$

$S \to (S)$

$S \to a$

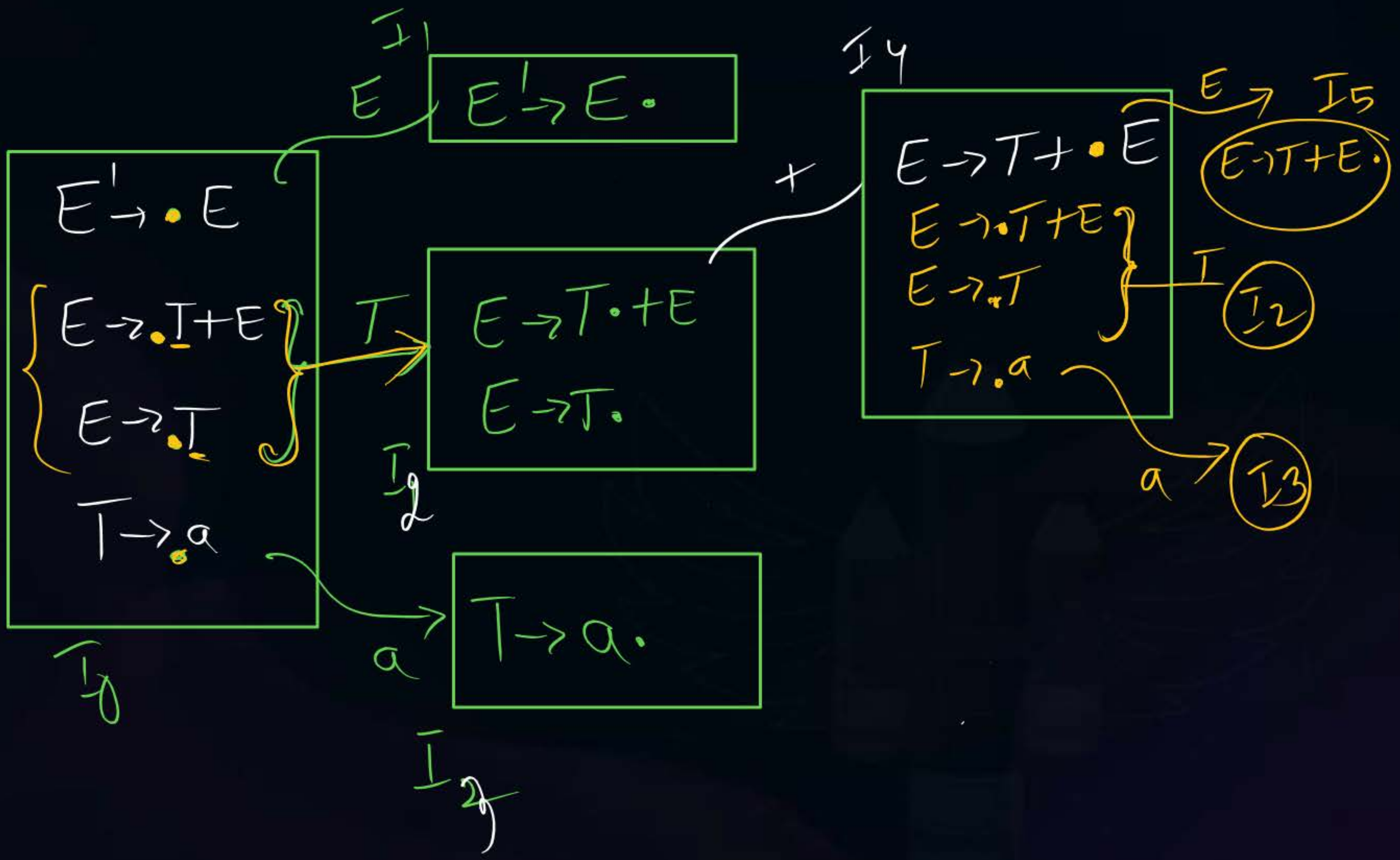


$I_0 \cdots I_5$

6 States

(Q) How many states present in LR(0) items DFA for given grammar

$E' \to E$
$E \to T + E$
$E \to T$
$T \to a$



6 states

LR(0) items DFA states ??

$$\left\{\begin{array}{l} S \to Aa \\ S \to Bb \\ A \to a \\ B \to b \end{array}\right\}$$

2 mins Summary

Topic One

Topic Two

Topic Three

Topic Four

Topic Five

THANK - YOU